

# BLOCKCHAIN – ENABLED DRUG AUTHENTICATION AND TRACKING SYSTEM

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## ABSTRACT

The pharmaceutical supply chain has become highly complex due to the involvement of multiple stakeholders such as manufacturers, distributors, and pharmacies. This complexity has led to serious challenges including counterfeit drugs, lack of transparency, and inefficient tracking mechanisms. Counterfeit medicines not only pose a major threat to patient safety but also result in financial losses and reduced trust in healthcare systems. Traditional supply chain systems rely on centralized databases, which are highly vulnerable to data tampering, cyber-attacks, and unauthorized access. These limitations highlight the urgent need for a more secure and reliable solution.

To address these issues, this paper proposes a Blockchain–Enabled Drug Authentication and Tracking System that leverages decentralized technology to ensure secure and transparent drug tracking. The system records all transactions in an immutable blockchain, preventing any modification of data once it is stored. Smart contracts are used to automate key processes such as drug registration, shipment updates, and verification, thereby reducing manual effort and increasing efficiency.

Additionally, the system integrates IPFS for decentralized storage of drug-related documents, ensuring high data security and availability. The proposed solution improves transparency, prevents counterfeit drugs, and enhances trust among all stakeholders in the pharmaceutical supply chain.

## 1. INTRODUCTION

The pharmaceutical supply chain is a complex and multi-layered system that involves various stakeholders such as manufacturers, distributors, wholesalers, pharmacies, and regulatory authorities. Each participant plays a crucial role in ensuring that medicines are safely produced, transported, and delivered to end consumers. However, due to the involvement of multiple entities and processes, maintaining transparency, security, and accountability becomes a significant challenge. In traditional systems, data is usually stored in centralized databases, which makes them vulnerable to cyber-attacks, unauthorized access, and data manipulation.

One of the most critical issues faced by the pharmaceutical industry today is the increasing presence of counterfeit drugs. These fake medicines may contain incorrect or harmful ingredients, leading to severe health risks and even loss of life. The lack of proper tracking mechanisms makes it difficult to identify the origin of such drugs and remove them from the supply chain. Additionally, the absence of real-time tracking reduces visibility, making it harder for stakeholders to monitor the movement of drugs effectively.

Blockchain technology has emerged as a powerful solution to address these challenges by providing a decentralized and tamper-proof system for data management. It ensures that all transactions are

recorded in an immutable ledger, which cannot be altered once the data is stored. This increases transparency and trust among stakeholders, as every participant can verify the authenticity of information. Smart contracts further enhance the system by automating processes such as drug registration, shipment updates, and verification, reducing manual effort and errors.

In this project, a Blockchain-Enabled Drug Authentication and Tracking System is proposed to ensure secure, transparent, and efficient drug tracking across the entire supply chain. The system integrates blockchain with IPFS (InterPlanetary File System) for decentralized storage of drug-related documents. By implementing this approach, the system aims to eliminate counterfeit drugs, improve supply chain efficiency, and enhance trust among all participants.

## 2. PROBLEM STATEMENT

The pharmaceutical industry faces significant challenges due to the increasing presence of counterfeit drugs and the lack of transparency in traditional supply chain systems. Counterfeit medicines can enter the supply chain at various stages, making it difficult to detect and eliminate them effectively. These fake drugs may contain harmful or incorrect ingredients, posing serious health risks to patients and reducing confidence in healthcare services.

Another major issue is the reliance on centralized systems for storing and managing data. Such systems are highly vulnerable to cyber-attacks, data breaches, and manipulation, which can compromise the integrity of critical information. Additionally, stakeholders such as distributors and pharmacies often lack access to real-time data, making it difficult to verify the authenticity and origin of drugs. These challenges highlight the need for a secure, transparent, and decentralized system that ensures reliable tracking and verification of pharmaceutical products.

- Counterfeit drugs risk patient safety and public health
- No proper end-to-end tracking mechanism
- Centralized systems are insecure and vulnerable
- Difficult for stakeholders to verify authenticity
- Lack of transparency reduces trust

## 3. OBJECTIVES

The main objective of this project is to develop a blockchain-based system that ensures secure and transparent tracking of pharmaceutical products across the supply chain. The system aims to eliminate the problem of counterfeit drugs by maintaining an immutable and tamper-proof record of all transactions. It also focuses on improving the efficiency and reliability of supply chain operations through automation and real-time tracking.

Furthermore, the system provides a secure method for storing drug-related documents using decentralized technologies such as IPFS. It enables stakeholders including manufacturers, distributors, pharmacies, and regulatory authorities to easily verify the authenticity of drugs. By achieving these objectives, the proposed system enhances trust, transparency, and overall performance of the pharmaceutical supply chain.

- Develop blockchain-based tracking system
- Ensure transparency and traceability
- Prevent counterfeit medicines
- Enable drug verification by stakeholders
- Secure document storage using IPFS
- Improve efficiency and reliability
- Automate processes using smart contracts
- Provide real-time status updates across supply chain
- Ensure data immutability and tamper-proof records

#### 4. PROPOSED METHODOLOGY

The proposed system follows a structured and secure approach for tracking and verifying drugs throughout the supply chain. Initially, manufacturers register drug batches on the blockchain by providing essential details such as drug name, batch number, manufacturing date, and expiry date. This information is permanently stored on the blockchain, ensuring immutability and transparency.

Drug-related documents such as certificates and reports are uploaded to IPFS, and the generated hash values are stored on the blockchain. This ensures that documents are securely stored and can be verified at any time without risk of tampering. As drugs move through the supply chain, distributors update shipment status at each stage, and these updates are recorded on the blockchain, enabling real-time tracking.

Finally, pharmacies verify the authenticity of drugs by accessing blockchain records before selling them to consumers. This ensures that only genuine medicines reach the end users, improving safety and trust in the system.

- Drug batch registration by manufacturers
- Immutable storage using blockchain
- IPFS document upload and hash storage
- Shipment updates by distributors
- Real-time tracking of drug movement
- Verification by pharmacies before sale

#### 5. SYSTEM ARCHITECTURE

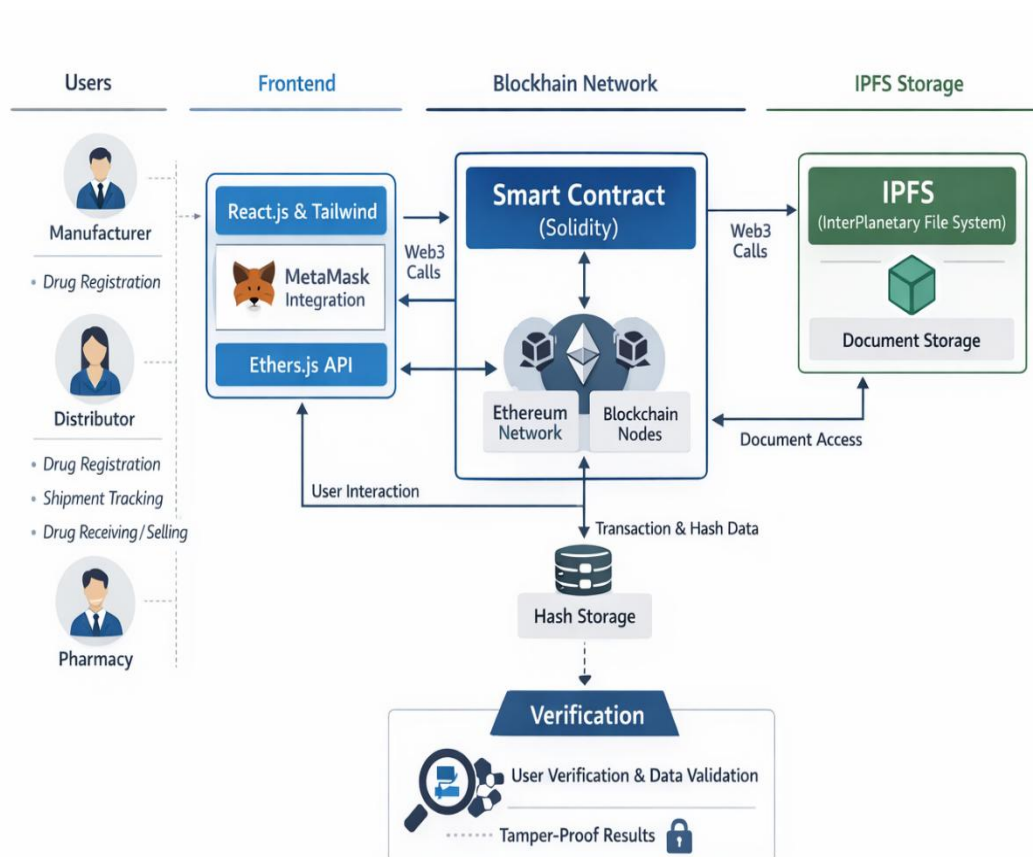
The system is designed using a multi-layer architecture that ensures secure and efficient operation. Each layer is responsible for specific functionalities and works together to provide a seamless experience. The frontend layer provides a user-friendly interface for interacting with the system, allowing users to register, log in, and perform various operations.

The backend layer handles business logic and ensures smooth communication between the frontend and the blockchain network. The blockchain layer is responsible for storing all transaction data securely using smart contracts, ensuring immutability and transparency. Additionally, the storage layer uses IPFS for storing large files such as drug-related documents in a decentralized manner.

MetaMask is integrated into the system to enable secure user authentication and interaction with the blockchain. This layered architecture ensures scalability, security, and efficient data flow across the system.

- Frontend using React.js for UI
- Backend using Node.js and Express.js
- Blockchain using Ethereum and Solidity
- IPFS for decentralized storage
- MetaMask for authentication
- Secure interaction between all layers

## SYSTEM ARCHITECTURE DIAGRAM



## 6. WORKING

The system operates in a well-defined sequence to ensure secure and transparent tracking of drugs. The process begins with user authentication, where users log in and connect their MetaMask wallet for secure access. Manufacturers play a key role by registering drug batches and uploading relevant documents to the system.

As the drugs move through the supply chain, distributors update the shipment status at each stage. These updates are recorded on the blockchain, ensuring that every transaction is transparent and traceable. Pharmacies then verify the authenticity of drugs by accessing blockchain records before selling them to consumers.

This step-by-step process ensures that only genuine drugs are delivered to end users, thereby improving safety, trust, and efficiency in the pharmaceutical supply chain.

Additionally, each transaction is time-stamped and permanently stored, making it easy to track the complete history of a drug batch. The use of smart contracts automates the process and reduces the need for manual intervention. The decentralized nature of blockchain eliminates the risk of a single point of failure. IPFS storage ensures that large documents are securely stored without overloading the blockchain. Real-time updates help stakeholders make quick and informed decisions. Overall, the system enhances accountability and minimizes the chances of fraud in the supply chain.

Furthermore, the system provides end-to-end visibility for all stakeholders involved. It reduces dependency on intermediaries, thereby lowering operational costs. Data integrity is maintained as records cannot be altered once stored on the blockchain. The system can be easily scaled to handle large volumes of transactions. It also supports quick verification during audits and inspections. User roles and permissions ensure that only authorized actions are performed. This structured workflow improves overall coordination across the supply chain.

- User login and wallet connection
- Manufacturer registers drug details
- Documents uploaded to IPFS
- Distributor updates shipment status
- Blockchain records all transactions
- Pharmacy verifies before selling

## WORKING FLOW DIAGRAM





## 7. ADVANTAGES

The proposed system offers several advantages that significantly improve the pharmaceutical supply chain. It ensures a high level of security by storing data in a tamper-proof blockchain environment, making it impossible to alter or delete records. The system also provides complete transparency, allowing all stakeholders to access and verify transaction data.

By automating processes using smart contracts, the system reduces manual effort and minimizes errors. It also enables real-time tracking of drug movement, improving efficiency and reliability. Overall, the system enhances trust among stakeholders and ensures the delivery of safe and authentic medicines.

- Prevents counterfeit drugs
- Tamper-proof data storage
- Improves transparency
- Builds trust among stakeholders
- Enables real-time tracking
- Reduces manual errors

## 8. CONCLUSION

The Blockchain-Enabled Drug Authentication and Tracking System provides a secure and efficient solution for managing pharmaceutical supply chains. By leveraging blockchain technology, the system ensures that all transactions are recorded in a transparent and immutable manner, preventing data manipulation and fraud.

The integration of IPFS further enhances data security by providing decentralized storage for drug-related documents. This combination of technologies ensures that the system is reliable, scalable, and resistant to cyber threats. The proposed system not only prevents counterfeit drugs but also improves trust, efficiency, and transparency across the supply chain.

In the future, this system can be expanded with advanced technologies such as IoT and AI to further enhance its capabilities and effectiveness.

- Ensures secure and transparent tracking
- Prevents counterfeit drugs
- Enhances trust and reliability

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